

CHAPTER 6

SANITATION



CHAPTER 6

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CHAPTER 6

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SECTION 6-1

SANITATION SCHEDULE

⚠ DANGER

Electrical shock possible. Disconnect and lockout power before cleaning. See **Section 1-3, Zero Energy State Procedures**.

⚠ DANGER

Running or coasting mixing arms can cause serious injury or death. Never reach inside mixing bowl without disconnecting and locking out power.

⚠ CAUTION

Make sure all covers, guards and other parts that are removed for cleaning are re-attached and secured before running mixer.

⚠ CAUTION

Soaps with high Ph values can cause damage to inside bowl surface over a period of time. Use mild soap to clean mixer.

Your **Peerless** mixer has been designed and built for easy cleaning. We recommend the following cleaning schedule and instructions to avoid unsanitary conditions and to add to the life of your mixer.

Sometimes personal experience dictates a sanitation schedule different from our recommendation. However, it is *important* that you develop and enforce a thorough, periodic cleaning program that covers your mixer's sanitation needs.

Follow all safety and sanitation guidelines when working on the mixer. Before starting to clean your mixer, always disconnect and lock out electrical power. See **Section 1-3, Zero Energy State Procedures**.

We strongly recommend thorough cleaning of your new mixer before you mix the first batch.

SECTION 6-1

SANITATION SCHEDULE TABLE 6-1-1

⚠ DANGER Avoid personal injury or death. Disconnect and lockout power before cleaning. See **Section 1-3, Zero Energy State Procedures**.

SUGGESTED AREA	PROCEDURE	FREQUENCY
Agitator Shaft Seals	Disassemble, clean with warm, soapy water. Pack with food-grade lubricant.	Daily (each down day)
Canopy Scrapers	Remove dough build-up with warm, soapy water or scrape with plastic scraper.	Weekly
Drive Motor	Clean fan cover. Dry clean housing with brush.	Monthly
Flour Dust Vent/Cage Sock	Remove excess flour, dough.	Weekly
Flour Gate	Remove flour and dough with soapy water.	Weekly
Front/Rear Bowl Seals	Remove dough build-up.	Weekly
Inside Bowl	Clean with hose and plastic scraper or machine-wash.	Daily (each down day)
Mixer End Columns	Wipe out flour dust or oil.	Monthly
Mixer Outside	Clean with warm, soapy water.	Daily (each down day)
Safety Ingredient Door	Clean frame, underside of door.	Daily (each down day)

SECTION 6-1

DAILY CLEANING DETAIL 6-1-1

We recommend daily cleaning of the following areas of your **Peerless** mixer. If your mixer operates on a 24-hour schedule, and daily cleaning is not possible, we recommend cleaning during every scheduled nonproduction, or down time. Clean by washing down with water or by scraping and wiping the surfaces.

We also recommend daily cleaning of any other surfaces that come into direct contact with dough.

⚠ DANGER

Avoid personal injury or death. Disconnect and lockout power before cleaning. See **Section 1-3, Zero Energy State Procedures**.

⚠ WARNING

When spraying mixer or mixer components with water or compressed air, always wear eye protection to prevent objects entering eyes.

DAILY CLEANING:

⚠ CAUTION

Failure to clean agitator shaft seals can cause damage to agitator shaft.

Agitator Shaft Seals.

- ✓ Disassemble and clean with warm, soapy water. See **Section 6-2**.
- ✓ Pack **seals** with Petrol Gel, solid shortening, or equivalent food-grade lubricant.

SECTION 6-1

DAILY CLEANING (*con't.*):

 CAUTION

Avoid personal injury and damage to mixer bowl. Be sure refrigeration system is turned off before cleaning **inside bowl**.

Inside Bowl.

- ✓ We recommend washing with a hose and plastic scraper for **inside-bowl** cleaning. See **Section 6-5, Inside Bowl**.
- ✓ If machine-wash method is used, fill bowl to agitator shaft with soap and water (and optional sanitizing agent). Run 30 minutes in **low speed**. See **Section 6-5, Inside Bowl**.
- ✓ If washing **inside bowl** with a high-pressure hose, use only water. We do not recommend washing **inside bowl** with a high-pressure hose if very hot water and/or powerful sanitizing chemicals are used. See **Section 6-5, Inside Bowl**.
- ✓ We do not recommend steam cleaning **inside bowl**.

Mixer Outside.

- ✓ Clean with warm, soapy water.

NOTE:

Do not spray the operator control panel, or any electrical covers or conduits, directly with water or other liquids.

SECTION 6-1

WEEKLY CLEANING DETAIL 6-1-2

We recommend the following areas of your **Peerless** mixer be cleaned thoroughly once a week.

▲ DANGER

Avoid personal injury or death. Disconnect and lockout power before cleaning. See **Section 1-3, Zero Energy State Procedures**.

WEEKLY CLEANING:

**Canopy Scrapers
(Optional).**

- ✓ Check for dough buildup. Clean with warm, soapy water or scrape off dough with plastic scraper.

**Flour Dust Vent
Cage/Sock.**

- ✓ Remove sock from cage. Remove excess flour from sock. If using compressed air to blow sock clean, wear safety glasses for eye protection. See **Section 6-4, Flour Dust Vent Cage/Sock**.
- ✓ Clean cage to remove flour and dough.

▲ CAUTION

Avoid personal injury. Be sure pneumatic system is turned off and locked out before cleaning **flour gates** or **safety ingredient doors**. See **Section 1-3, Zero Energy State Procedures**.

Flour Gate.

- ✓ Disassemble and clean with warm, soapy water to remove flour dust and dough. See **Figure 6-3-1 Flour Gate Assembly**.

Front/Rear Bowl Seals.

- ✓ Check for dough buildup. Remove; scrape clean.

Safety Ingredient Door.

- ✓ Check for dough/gluten buildup on underside of door. Wipe out any materials present.
- ✓ Clean flour and dough from frame.

SECTION 6-1**MONTHLY CLEANING DETAIL 6-1-3**

We recommend the following areas of your **Peerless** mixer be cleaned once a month.

▲ DANGER

Avoid personal injury or death. Disconnect and lockout power before cleaning. See **Section 1-3, Zero Energy State Procedures**.

MONTHLY MAINTENANCE:

Drive Motor.

- ✓ Remove and clean **drive-motor** fan cover.
- ✓ Dry clean **drive-motor** housing with a brush. Be sure ribs (veins) are free of flour, dough or other debris.

Mixer End Columns.

- ✓ Clean by wiping out any flour dust or oil that may be present.
Do not wash inside end doors with water.

SECTION 6-2

AGITATOR SHAFT SEAL CLEANING

▲ DANGER

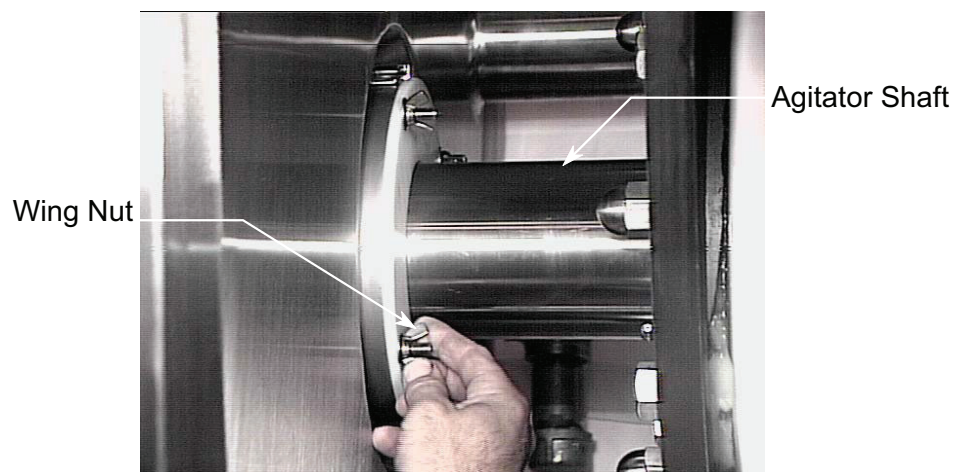
Cleaning mixer while power is on could cause serious injury or death. Avoid accidental operation of mixer. Lockout electrical power before cleaning. See **Section 1-3, Zero Energy State Procedures**.

The **agitator shaft seals** on your **Peerless** mixer are designed for quick and easy cleaning. **Peerless** recommends the **agitator shaft seals** be cleaned daily or during every scheduled nonproduction, or down time. We realize personal experience may dictate a cleaning schedule different from our recommendation; however, shaft seals should be cleaned at least once or twice a week to prevent damage to the seals and/or agitator shaft. Follow all safety guidelines and OSHA rules when cleaning the mixer.

We recommend the following procedure for cleaning agitator shaft seals.

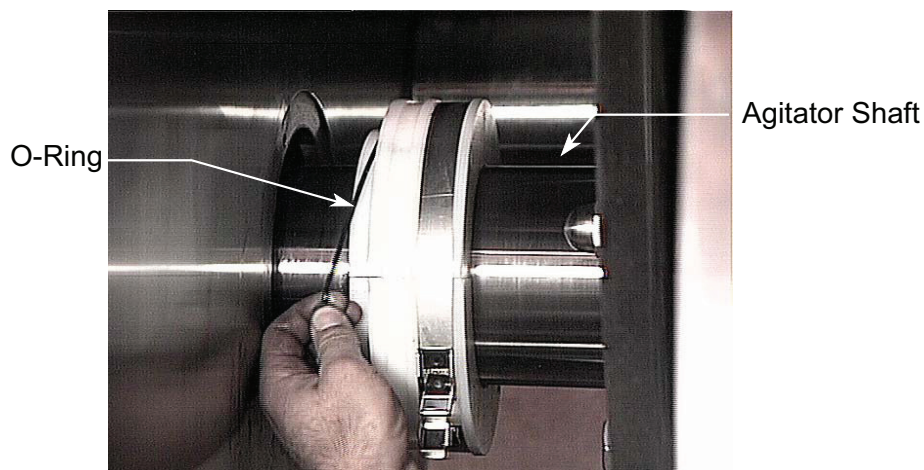
TO CLEAN AGITATOR SHAFT SEALS:

1. Disconnect and lockout electrical power to mixer. See **Section 1-3, Zero Energy State Procedures**.
2. Remove wing nuts that hold **shaft seal assembly** to bowl ends.

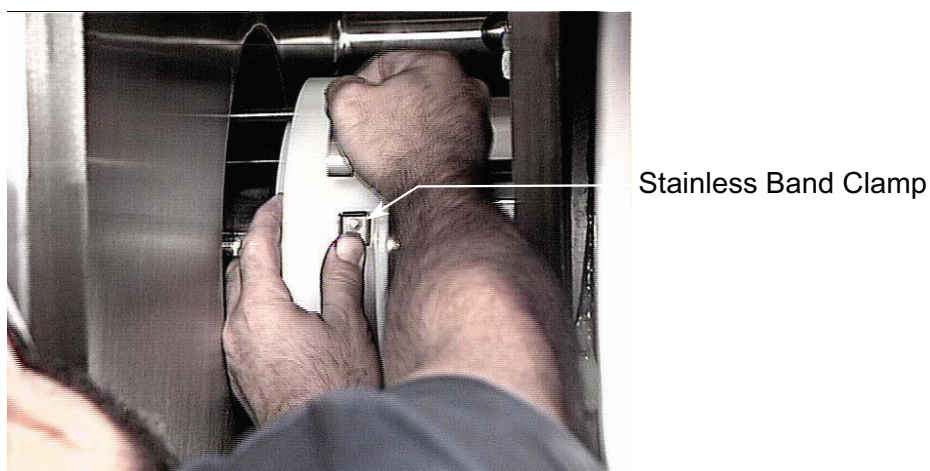


SECTION 6-2

3. Slide **shaft seal assembly** away from bowl end. Remove O-ring from **shaft seal assembly**.

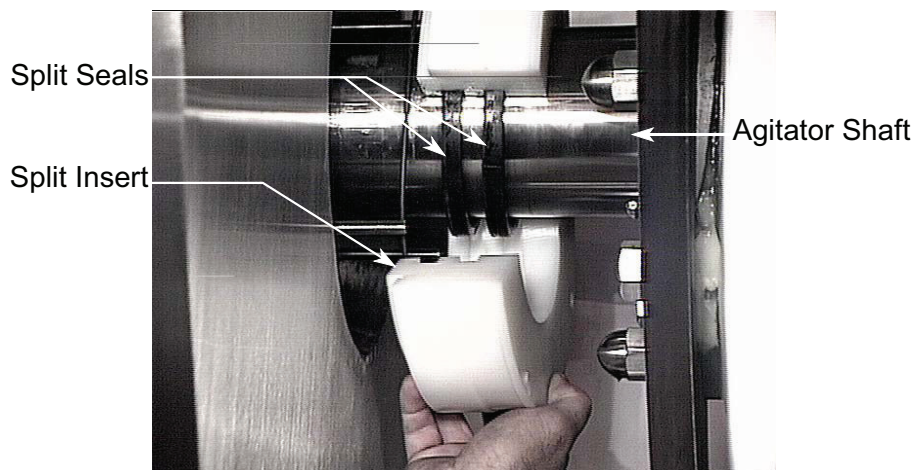


4. Unsnap and remove stainless band clamp.

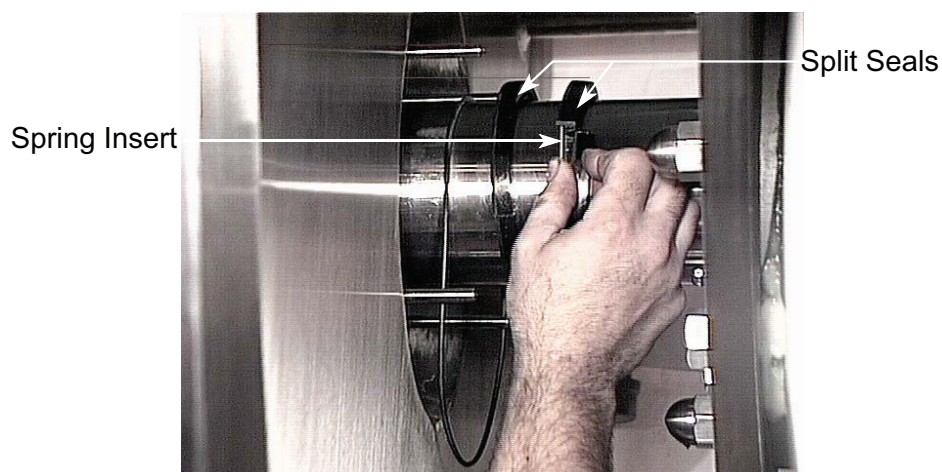


SECTION 6-2

5. Pull apart plastic split inserts to expose split seals.



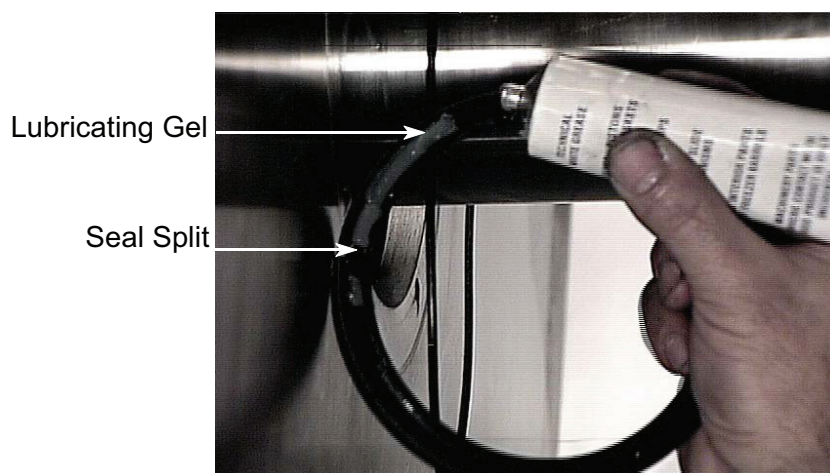
6. Remove spring insert from split seals. *It is best to not unhook the spring.* Use a finger to slide the spring out of the lip of the seal. Let the spring stay on the shaft.



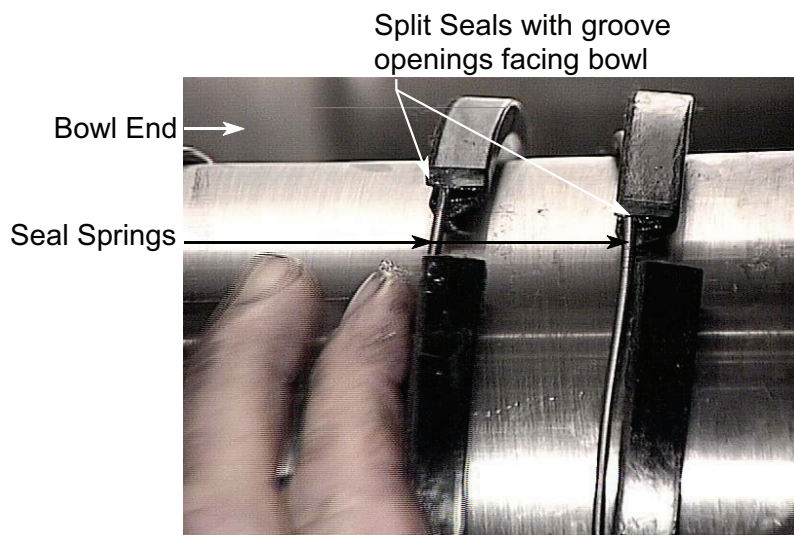
7. Inspect seals. Replace if hard and brittle, or show excessive wear.
8. Clean seals and seal housing by washing in warm water.

SECTION 6-2

9. Lubricate split seals and seal inserts with Petrol Gel, white petroleum jelly or shortening to help seal and prolong the life of the seals.

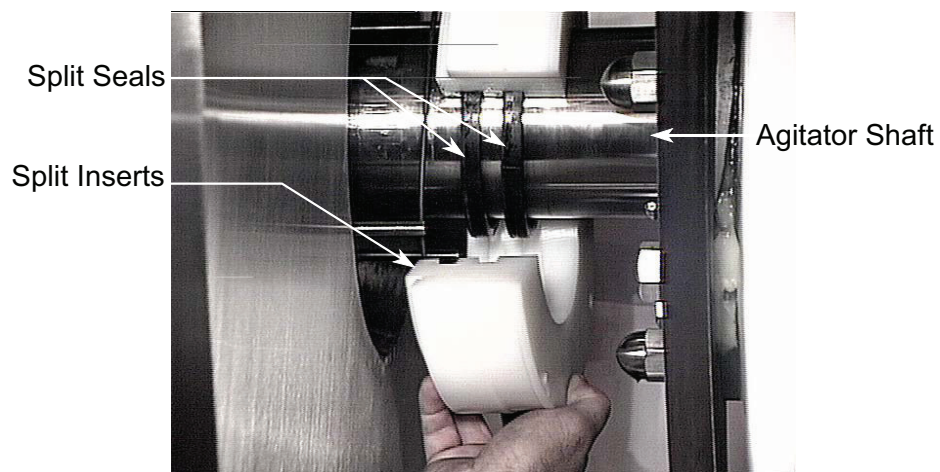


10. Reinstall split seals on shaft making sure groove openings face mixer bowl. Guide seal spring into grooves. Grease seals and fill grooves with Petrol Gel, white petroleum jelly or shortening.

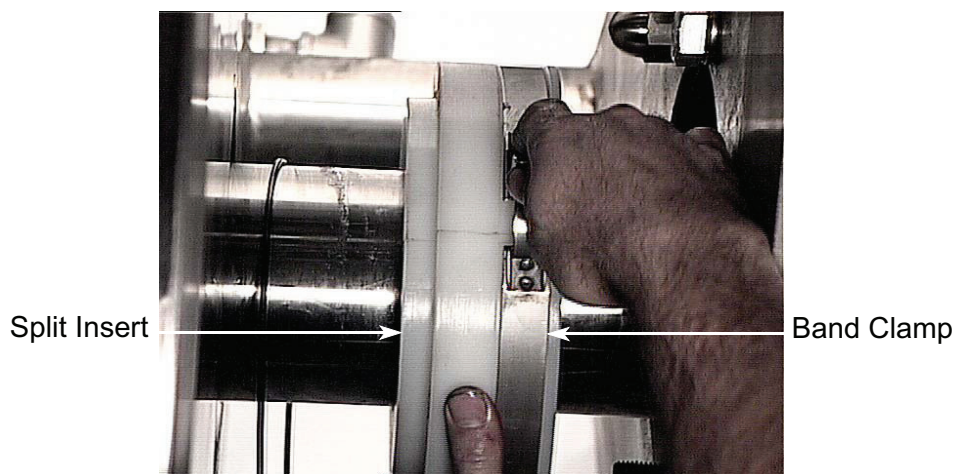


SECTION 6-2

11. Reinstall split inserts over split seals on shaft.

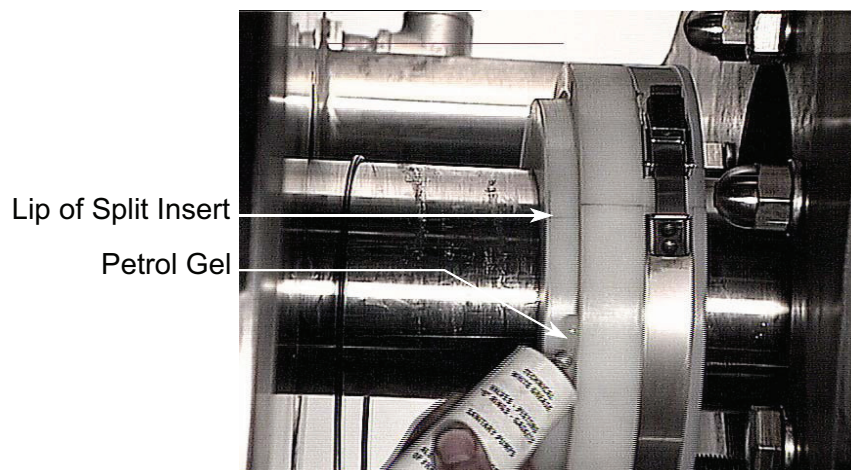


12. Place band clamp over split insert and snap shut.

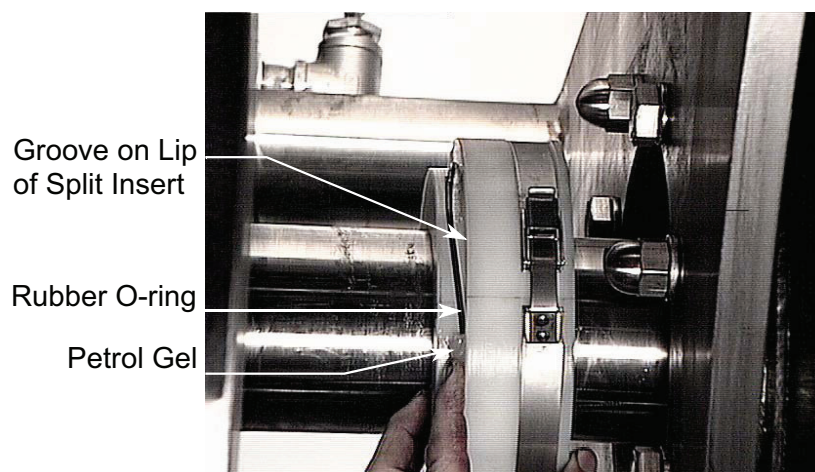


SECTION 6-2

13. Install a bead of Petrol Gel in O-ring groove on lip of split insert.

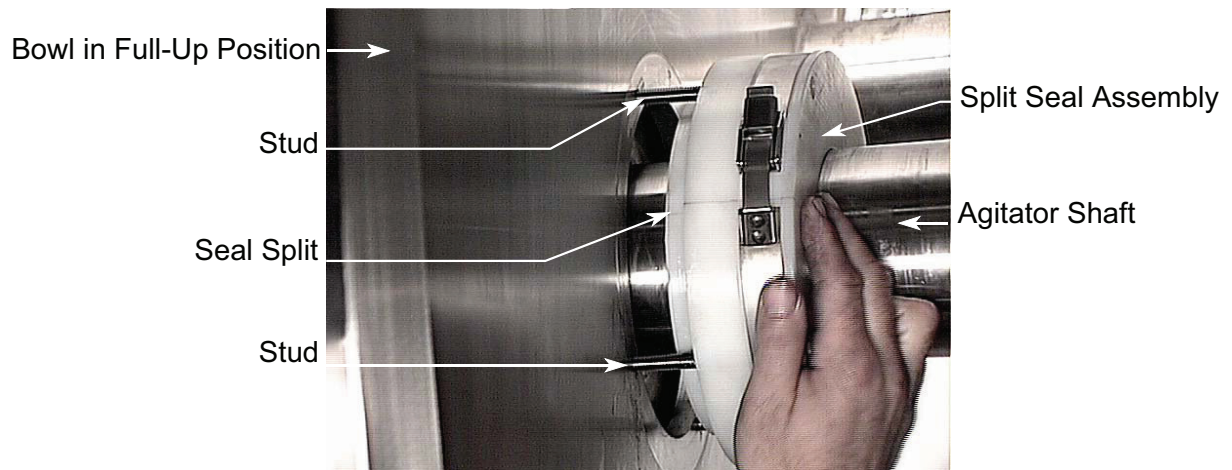


14. Install rubber O-ring in groove on lip of split insert.



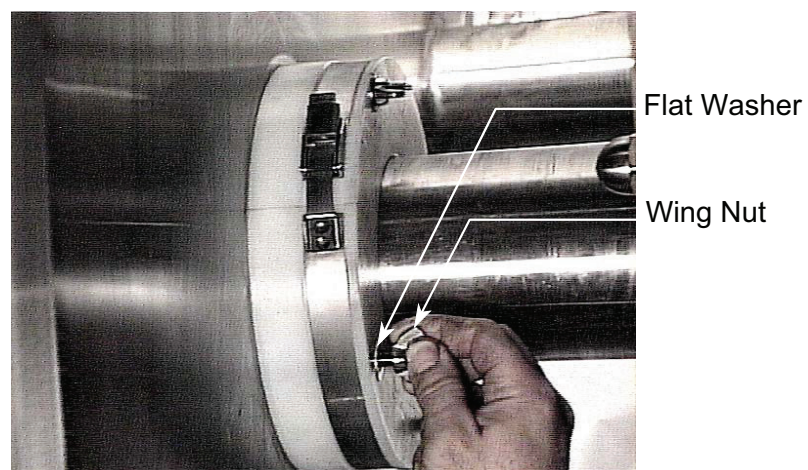
SECTION 6-2

15. Slide split seal assembly over stud mounting holes. *Be sure seal splits between housing halves are horizontal when bowl is in full-up position.*



NOTE: In some applications, studs and wing nuts are welded together. Insert studs through shaft seal assembly and into bowl end. Secure by tightening wing nuts.

16. Install flat washers and wing nuts on studs. Tighten wing nuts.



SECTION 6-3

FLOUR GATE CLEANING

⚠ DANGER

Severe personal injury or death may result from improper contact with **flour gate slide**. Turn off air supply, bleed off air pressure and lockout before cleaning. See **Section 1-3, Zero Energy State Procedures**.

⚠ DANGER

Do not walk on top of mixer. Surfaces of mixer may be slippery and could cause fall resulting in injury.

⚠ CAUTION

Do not reach hand or arm into flour gate opening. Severe personal injury may result.

The **flour gate** on your **Peerless** mixer is designed for easy cleaning. Flour dust, dough and gluten can get into **flour gate** openings and fittings, causing unsanitary conditions and less efficient operation of your mixer.

Check **Chapter 2, Options Checklist**, to see which type **flour gate** has been installed on your mixer:

- ✓ **Standard Butterfly-style Valve**
- ✓ **Optional Pneumatic Slide Gate.**

Peerless recommends that you keep on hand a spare flour dust vent sock and flour gate-to-flour hopper flexible connection, or sleeve for rotation purposes. Rotate the flour dust vent sock and customer-supplied sleeve weekly during sanitation. See **Section 6-4, Flour Dust Vent Cage/Sock**.

CLEANING AN OPTIONAL PNEUMATIC FLOUR SLIDE GATE

We recommend the following steps for cleaning a **pneumatic slide gate**.

1. Disconnect and lockout electrical power to the mixer. See **Section 1-3, Zero Energy State Procedures**.
2. Shut off the mixer's air supply by turning the mixer's shut-off valve located behind the end doors, in front of the air supply filter-regulator-coalescer assembly. When the shut-off valve is turned, the air pressure will bleed off.
3. Lock out the air supply to the mixer. See **Section 1-3, Zero Energy State Procedures**.
4. Remove the flour sock from between mixer and hopper.
5. Remove the retainer knobs from the top of the **flour gate** assembly.
6. Lift off the flour flange retainer plate exposing the **flour gate** assembly.
7. Wipe off the **flour gate slide**, retainer plate and air cylinder. Wash any plastic parts in cool water. **DO NOT USE HOT WATER.**
8. Reassemble **flour gate** assembly making sure to tighten the knobs securely.

CLEANING AN OPTIONAL BUTTERFLY-STYLE VALVE

We recommend the following steps for cleaning an optional **butterfly-style valve**.

1. Disconnect and lockout electrical power to the mixer. See **Section 1-3, Zero Energy State Procedures**.
2. Shut off the mixer's air supply by turning the mixer's shut-off valve located behind the end doors, in front of the air supply filter-regulator-coalescer assembly. When the shut-off valve is turned, the air pressure will bleed off.
3. Lock out the air supply to the mixer. See **Section 1-3, Zero Energy State Procedures**.
4. Remove flour sock from between mixer and hopper.
5. Lower bowl and scrape the bottom of the flour gate to clean.
6. Reinstall flour dust vent sock.

 CAUTION

Do not reach hand or arm inside butterfly valve opening when reconnecting air lines. Valve may move unexpectedly causing personal injury.

NOTE:

When reinstalling, be sure to replace proximity cables on correct proximity pickups.

SECTION 6-4

FLOUR DUST VENT CAGE/SOCK

▲ DANGER

Accidental operation of mixer while cleaning can cause severe personal injury or death. Disconnect and lockout power before cleaning. See **Section 1-3, Zero Energy State Procedures**.

The **flour dust vent sock** covers the **flour dust vent cage** to catch and contain flour dust particles. See **Figure 3-11-1, Flour Dust Vent Assembly**.

Peerless recommends that you keep a spare **flour dust vent sock** on hand. Remove the **vent sock** weekly from the **flour dust vent cage** and replace it with the spare **sock**. Machine wash the soiled **vent sock** and have it ready for the next sanitation rotation or in case problems occur with the **sock** in use. To order optional spare **flour dust vent socks**, call the Parts Department at **Peerless Machinery Corp**.

We recommend, also, that you keep a spare flour-gate to flour-hopper flexible connection, or sleeve. See **Figure 3-8-1, Flour Gate Connection**. Rotate the customer-supplied sleeve weekly during scheduled sanitation, the same as with the **flour dust vent sock**.

NOTE: It is important to have available a freshly sanitized sleeve and **flour dust vent sock** for every sanitation cycle.

SECTION 6-5

INSIDE BOWL

⚠ DANGER

Accidental operation of mixer while cleaning can cause severe personal injury or death. Disconnect and lockout power before cleaning. See **Section 1-3, Zero Energy State Procedures**.

⚠ CAUTION

If cleaning mixer by high-pressure hose-wash method, always use protective eye wear and rain gear.

We recommend daily cleaning inside the mixer's bowl. Clean inside the bowl by filling it with warm water and running the agitators in low speed for no more than 30 minutes. Remove water from the bowl by pouring into a trough or by draining through the optional bowl drain valve.

To use the optional bowl drain valve, attach a 2" ID hose to the valve discharge fitting. Loosen the valve hand knob by turning it counterclockwise. Pull the hand knob out to open the valve. *Do not tilt the bowl until the drain hose has been removed.*

⚠ CAUTION

During machine-wash, *do not exceed 30 minutes running time*, especially in **low speed**. Motor overheating can occur.

Caustic sanitizers are known to etch stainless steel interiors and seals over time, especially if used in excess of manufacturer's parts-per-million recommendations. It is important to monitor usages of your specific sanitizing chemical to minimize damage without compromising mixer sanitation requirements.

CAUTION

During sanitization, *always* remove agitator shaft seals, clean and relubricate with food grade lubricant. See **Section 6-2, Agitator Shaft Seal Cleaning**.

We do not recommend steam cleaning the **inside bowl**.

SECTION 6-6

MIXER END COLUMNS

 DANGER

Accidental operation of mixer while cleaning can cause severe personal injury or death. Disconnect and lockout power before cleaning. See **Section 1-3, Zero Energy State Procedures**.

 CAUTION

Do not wash inside the end doors with water. Components inside end enclosure are not corrosion proof.

Peerless recommends cleaning the **mixer end columns** once a month by wiping out excess grease and oil with a dry cloth. Flour dust can be wiped out with a dry cloth or vacuumed out.